

The Intermediate Schatten Gelfand-Number Gap

An explicit literature answer to arXiv:2011.06554

Automated mathematics run `fa_banach_001`, lane 14

22 July 2026

Status: literature_already_answered. This note records an exact later-literature answer, not a new result.

Original question

Hinrichs, Prochno, and Vybíral study the Gelfand numbers of the identity

$$c_n(\mathcal{S}_p^N \hookrightarrow \mathcal{S}_q^N), \quad 1 \leq p \leq 2 \leq q \leq \infty.$$

In Section 4.6, p. 21 of the arXiv PDF, they state that their upper and lower bounds match for small and very large codimension, and write:

“We leave it an open problem to find a good lower bound also in the intermediate range $c_{p,q}N^2 \leq n \leq N^2 - N_q + 1$,” where $N_q = cN^{1+2/q}$.

This is the exact signal extracted from arXiv:2011.06554 [1].

Explicit answering paper

Prochno and Strzelecki [2] explicitly identify the connection. On p. 5 they say that their first Gelfand-number result “closes a gap” in the 2020 paper and that the intermediate-codimension upper bounds there are sharp. Their Theorem A (p. 6) and Proposition 3.1 (p. 19) prove, for $0 < p \leq 2 \leq q \leq \infty$,

$$c_n(\mathcal{S}_p^N \hookrightarrow \mathcal{S}_q^N) \gtrsim N^{-1/2-1/p} \sqrt{N^2 - n + 1}$$

whenever

$$(1 - c)N^2 \leq n \leq N^2 - cN^{1+2/q} + 1.$$

This matches the upper bound in the original paper. For $1 \leq p \leq 2$, their alternative proof gives lower-bound constant 1.

Proposition 3.2 (p. 20) treats the complementary range:

$$c_n(\mathcal{S}_p^N \hookrightarrow \mathcal{S}_q^N) \gtrsim \min \left\{ 1, \frac{N^{3/2-1/p}}{\sqrt{n}} \right\}, \quad 1 \leq n \leq (1 - c)N^2.$$

Together Propositions 3.1 and 3.2 cover the full intermediate strip in the source question and give the matching lower asymptotics sought there.

Identification and scope

This is an exact `literature_already_answered` match: the supporting authors cite the original paper, call out its gap, and prove the matching lower bounds. No portion is promoted as an original proof. The identification is limited to the source's $1 \leq p \leq 2 \leq q \leq \infty$ intermediate-range question; separate conjectures for $2 \leq p \leq q$ and quasi-Banach regimes in the supporting paper are outside this packet.

References

- [1] A. Hinrichs, J. Prochno, and J. Vybíral, *Gelfand numbers of embeddings of Schatten classes*, arXiv:2011.06554 (2020), Section 4.6, p. 21.
- [2] J. Prochno and M. Strzelecki, *Approximation, Gelfand, and Kolmogorov numbers of Schatten class embeddings*, arXiv:2103.13050 (2021), Theorem A and Propositions 3.1–3.2.